GSE2034 Data

1. Environment Setup & Data Loading

First, we load the necessary packages and the GSE2034 breast cancer dataset. Note that the filenames have been mapped so that Group 1 represents the 179 Relapse-Free patients, and Group 2 represents the 107 Metastasis patients.

```
library(highmean)
library(MASS)
library(ggplot2)
library(pheatmap)
library(moments)
library(reshape2)

normal_raw <- read.table("GSE2034-Tumor.txt", header = TRUE, row.names = NULL, sep = "\t", stringsAsFac
tumor_raw <- read.table("GSE2034-Normal.txt", header = TRUE, row.names = NULL, sep = "\t", stringsAsFac

X <- t(normal_raw[ , -1])
Y <- t(tumor_raw[ , -1])

X <- matrix(as.numeric(X), nrow = nrow(X), ncol = ncol(X))
Y <- matrix(as.numeric(Y), nrow = nrow(Y), ncol = ncol(Y))

n1 <- nrow(X)
n2 <- nrow(Y)
p <- ncol(X)

cat("Group 1 (Relapse-free) size:", n1, "\n")
```

```
## Group 1 (Relapse-free) size: 179
```

```
cat("Group 2 (Metastasis) size:", n2, "\n")
```

```
## Group 2 (Metastasis) size: 107
```

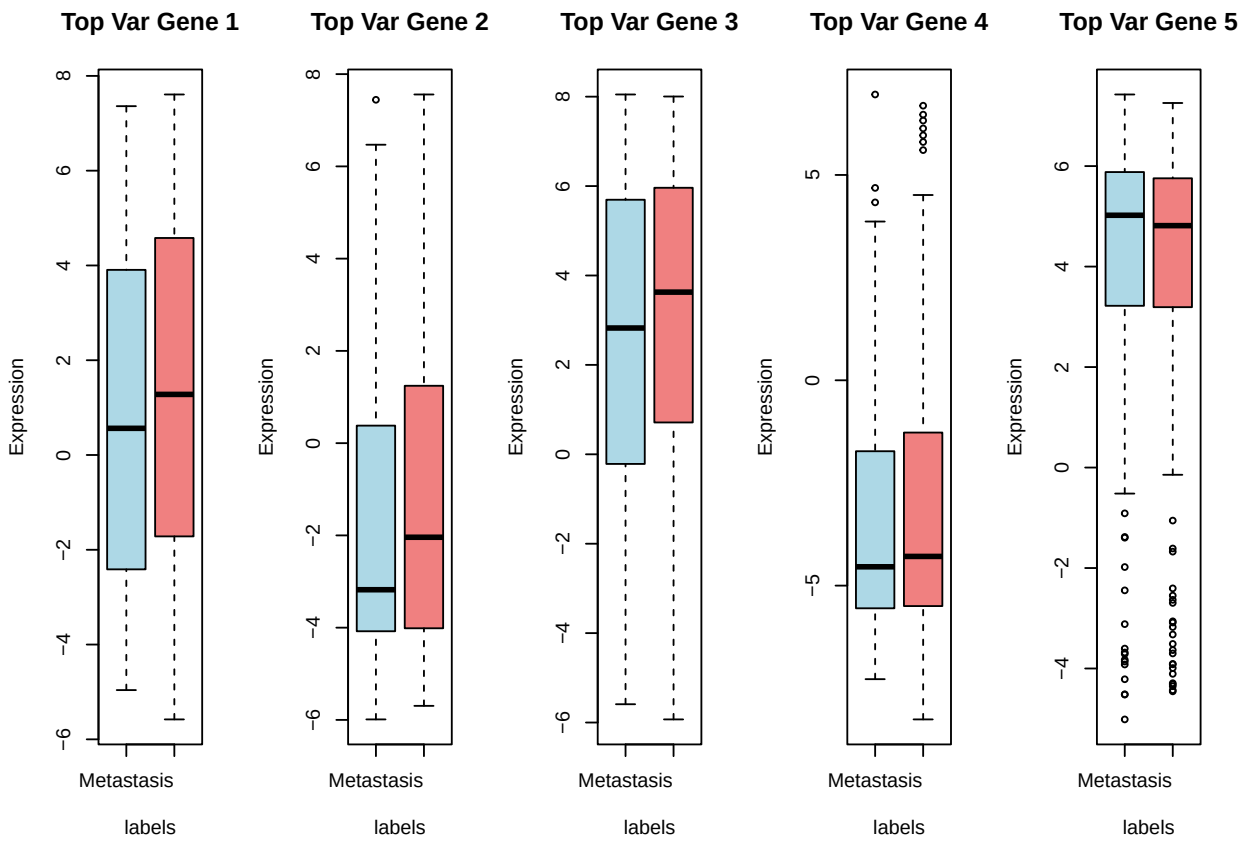
```
cat("Number of features (genes):", p, "\n")
```

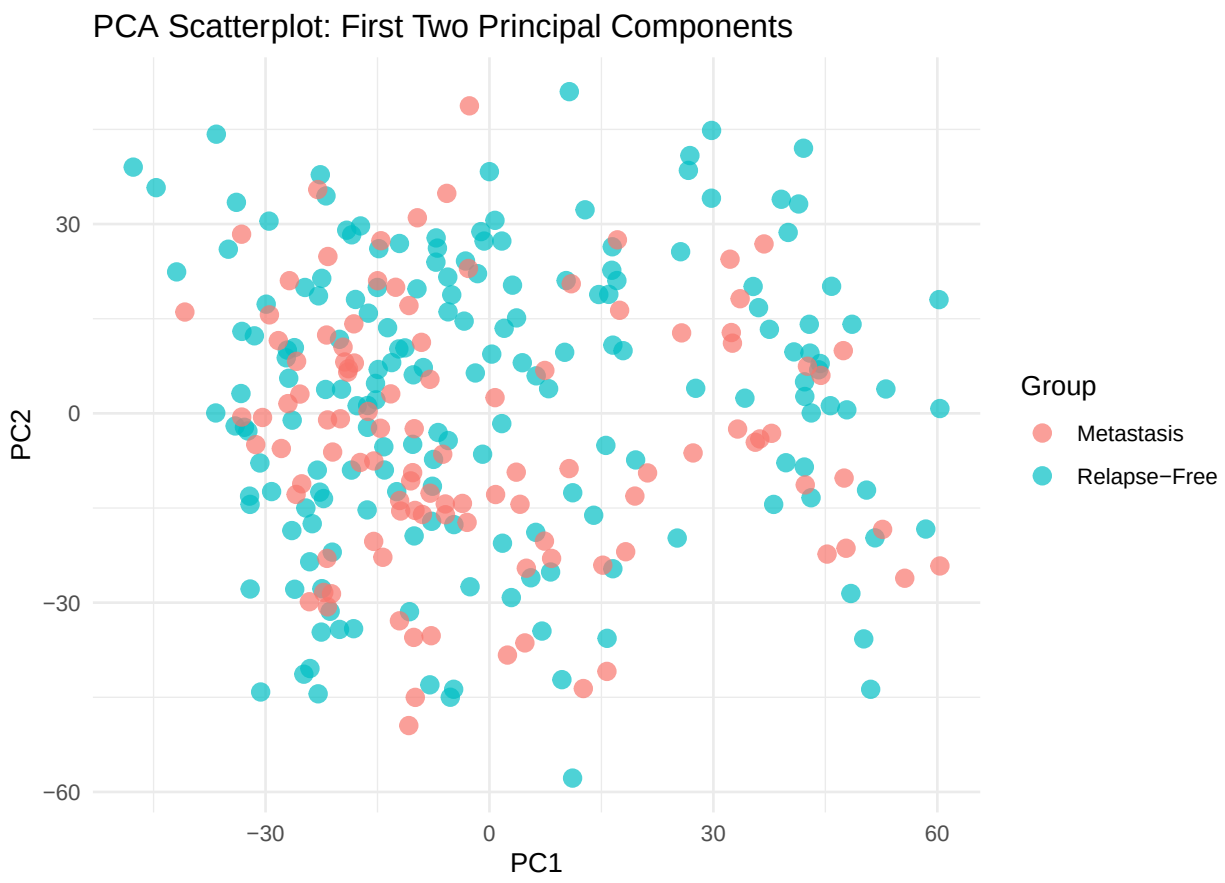
```
## Number of features (genes): 12634
```

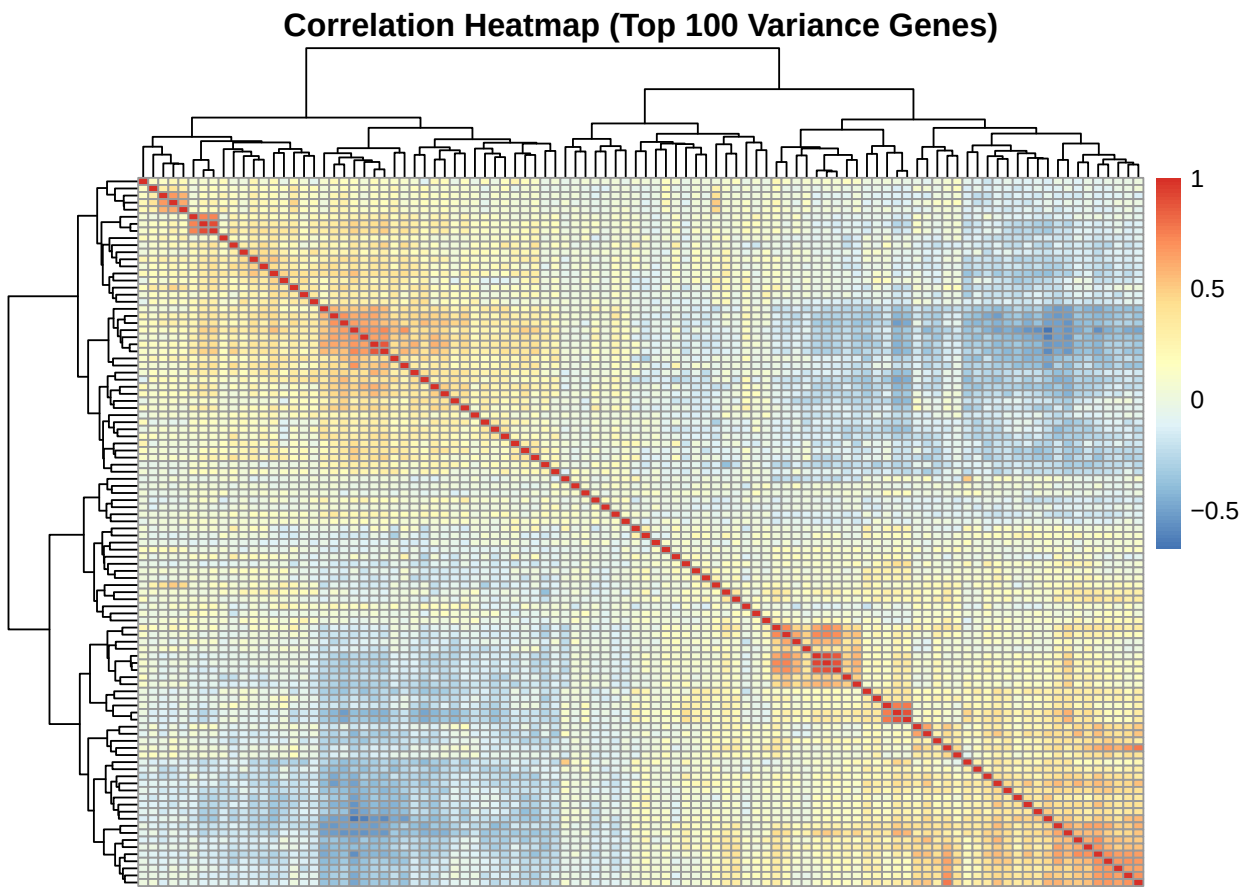
```
full_data <- rbind(X, Y)
labels <- c(rep("Relapse-Free", n1), rep("Metastasis", n2))
```

2. Exploratory Data Analysis (EDA)

```r
gene_vars <- apply(full_data, 2, var)
mean_var <- mean(gene_vars)
median_var <- median(gene_vars)

top_100_idx <- order(gene_vars, decreasing = TRUE)[1:100]
top_100_genes <- full_data[, top_100_idx]

par(mfrow=c(1,5))
for(i in 1:5){
  top_gene_idx <- order(gene_vars, decreasing = TRUE)[i]
  boxplot(full_data[, top_gene_idx] ~ labels,
          main = paste("Top Var Gene", i), ylab="Expression", col=c("lightblue", "lightcoral"))
}
par(mfrow=c(1,1))

pca_full <- prcomp(full_data, center = TRUE, scale. = FALSE)
df_pca <- data.frame(PC1 = pca_full$x[,1], PC2 = pca_full$x[,2], Group = labels)

ggplot(df_pca, aes(x = PC1, y = PC2, color = Group)) +
  geom_point(alpha = 0.7, size = 3) +
  theme_minimal() +
  ggtitle("PCA Scatterplot: First Two Principal Components")

cor_top100 <- cor(top_100_genes)
pheatmap(cor_top100, show_rownames = FALSE, show_colnames = FALSE,
         main = "Correlation Heatmap (Top 100 Variance Genes)")
```


Top Var Gene 1
Top Var Gene 2
Top Var Gene 3
Top Var Gene 4
Top Var Gene 5
Expression
Metastasis
labels


PCA Scatterplot: First Two Principal Components
PC2
30
0
−30
−60
−30
0
30
60
PC1
Group
Metastasis
Relapse−Free



Summary Metrics

```
balance_ratio <- n1 / n2
prop_var_top10 <- sum(pca_full$sdev[1:10]^2) / sum(pca_full$sdev^2)

cat("Balance Ratio (n1/n2):", round(balance_ratio, 3), "\n")
```

```
## Balance Ratio (n1/n2): 1.673
```

```
cat("Mean Feature Variance:", round(mean_var, 3), "\n")
```

```
## Mean Feature Variance: 0.709
```

```
cat("Median Feature Variance:", round(median_var, 3), "\n")
```

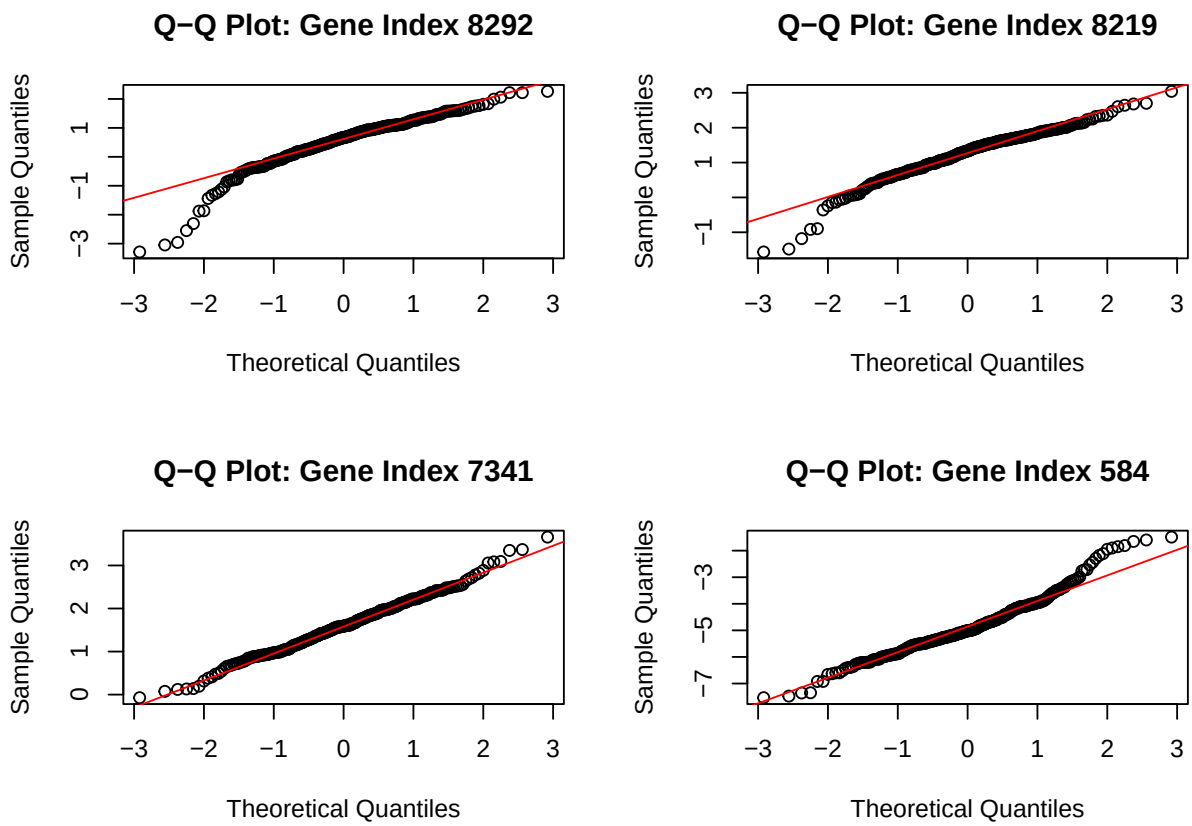
```
## Median Feature Variance: 0.459
```

```
cat("Variance Explained by Top 10 PCs:", round(prop_var_top10 * 100, 2), "%\n")
```

```
## Variance Explained by Top 10 PCs: 30.87 %
```

3. Assumption Checks

```
par(mfrow=c(2,2))
set.seed(2221)
sample_genes <- sample(1:p, 4)
for(g in sample_genes) {
  qqnorm(full_data[, g], main = paste("Q-Q Plot: Gene Index", g))
  qqline(full_data[, g], col="red")
}
```




```
par(mfrow=c(1,1))
```

4. Full Feature Set Testing (p = 12634)

```
get_p <- function(test_output) {
  if(is.list(test_output)) {
    if("p.value" %in% names(test_output)) return(as.numeric(test_output$p.value))
    if("pval" %in% names(test_output)) return(as.numeric(test_output$pval))
    return(as.numeric(tail(unlist(test_output), 1)))
  }
  return(as.numeric(test_output))
```

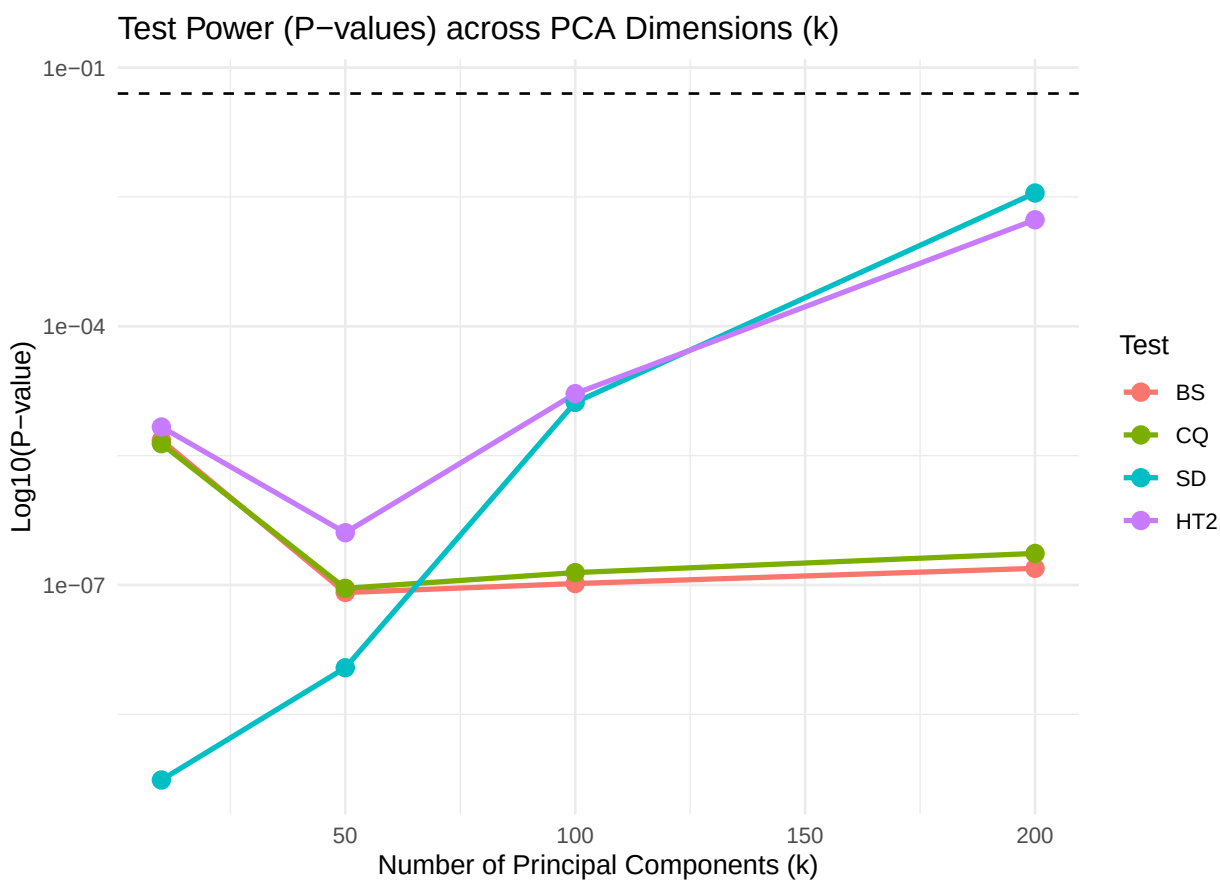
```r
}

pval_bs_full <- get_p(apval_Bai1996(X, Y))
pval_cq_full <- get_p(apval_Chen2010(X, Y))
pval_sd_full <- get_p(apval_Sri2008(X, Y))

data.frame(
  Test = c("Bai-Saranadasa", "Chen-Qin", "Srivastava-Du"),
  P_Value = c(pval_bs_full, pval_cq_full, pval_sd_full)
)
```

```
##             Test      P_Value
## 1 Bai-Saranadasa 2.898285e-07
## 2       Chen-Qin 3.514661e-07
## 3  Srivastava-Du 2.875133e-03
```

5. PCA Projection Analysis

```r
k_values <- c(10, 50, 100, 200)

hotellings_t2 <- function(X_proj, Y_proj) {
  n1 <- nrow(X_proj)
  n2 <- nrow(Y_proj)
  k <- ncol(X_proj)
  mean_diff <- colMeans(X_proj) - colMeans(Y_proj)

  S_pool <- ((n1 - 1) * cov(X_proj) + (n2 - 1) * cov(Y_proj)) / (n1 + n2 - 2)

  if (k < (n1 + n2 - 2)) {
    S_inv <- solve(S_pool)
    T2 <- (n1 * n2) / (n1 + n2) * as.numeric(t(mean_diff) %*% S_inv %*% mean_diff)
    F_stat <- (n1 + n2 - k - 1) / (k * (n1 + n2 - 2)) * T2
    p_val <- 1 - pf(F_stat, df1 = k, df2 = n1 + n2 - k - 1)
  } else {
    p_val <- NA
  }
  return(as.numeric(p_val))
}

results_pca <- data.frame(k = integer(), BS = numeric(), CQ = numeric(),
                          SD = numeric(), HT2 = numeric())

for (k in k_values) {
  X_k <- pca_full$x[1:n1, 1:k]
  Y_k <- pca_full$x[(n1 + 1):(n1 + n2), 1:k]

  p_bs <- get_p(apval_Bai1996(X_k, Y_k))
  p_cq <- get_p(apval_Chen2010(X_k, Y_k))
  p_sd <- get_p(apval_Sri2008(X_k, Y_k))
  p_ht2 <- hotellings_t2(X_k, Y_k)
```

```r
  results_pca <- rbind(results_pca, data.frame(k = k, BS = p_bs, CQ = p_cq,
                                               SD = p_sd, HT2 = p_ht2))
}

knitr::kable(results_pca, digits = 8)
```

k	BS	CQ	SD	HT2
10	4.78e-06	4.35e-06	0.00000000	0.00000679
50	8.00e-08	9.00e-08	0.00000001	0.00000040
100	1.00e-07	1.40e-07	0.00001309	0.00001654
200	1.60e-07	2.30e-07	0.00351189	0.00171806

```r
ggplot(melt(results_pca, id.vars = "k"), aes(x = k, y = value, color = variable)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  geom_hline(yintercept = 0.05, linetype = "dashed", color = "black") +
  scale_y_log10() +
  labs(title = "Test Power (P-values) across PCA Dimensions (k)",
       y = "Log10(P-value)", x = "Number of Principal Components (k)",
       color = "Test") +
  theme_minimal()
```